

Abdullah Shaukat

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EDUCATION

Drexel University

Philadelphia, PA

Bachelor of Science in Economics and Data Science

TECHNICAL SKILLS

Languages: Python, SQL, R, JavaScript, TypeScript, Visual Basic, C, HTML

Tools: Git, STATA, Oracle SQL, R-Studio, PyCharm, Jupyter Notebook, Power BI, Tableau, Excel, Power Query

Libraries: Pandas, NumPy, Matplotlib, Scikit-learn, NetworkX, Streamlit, Seaborn

ML/AI: NLP, Scikit-learn, Regression, TF-IDF, Sentence-BERT, PyTorch, TensorFlow, GenAI

EXPERIENCE

Data Engineering & Analysis Researcher

Apr 2026 – Present

Prof Zhiwei Chen, Drexel University

Philadelphia, PA

- Built a spatiotemporal simulator modeling flows for 1.1M+ daily journeys across a 147-station metro system
- Optimized shortest-path calculations, reducing unique path pairs from 1.1M to <22K
- Accelerated simulation runtimes by over 90% by eliminating redundant Dijkstra algorithm path executions
- Modeled trip time throughout the day ($R^2=0.76$) via linear regression, decomposing it into wait, speed, and dwell
- Identified critical gaps between topological network centrality (betweenness/degree) and actual ridership demand
- Quantified transfer-hub activity and commuter behavior across 6 card types, auto-generating station profiles

Operations Assistant

Sep 2025 – Mar 2026

Ronald McDonald House Charities

Philadelphia, PA

- Built Excel-based inventory tracking systems across 4+ operational areas, improving supply tracking by 30%
- Analyzed 5 years of family-stay data covering 1,000+ stays, building a dashboard that identified seasonal demand patterns and peak utilization periods to improve resource allocation and capacity planning
- Developed a donation analytics dashboard analyzing \$1M+ in historical contributions, identifying seasonal giving patterns and high-opportunity periods to better target fundraising and marketing campaigns
- Conducted facility-wide inventory audits and developed standardized Excel templates to improve spending visibility and reduce supply over-ordering by 20%

PROJECTS

ScentSync | *Python, Pandas, Scikit-learn, Sentence-BERT, Streamlit, NLP, Machine Learning* Mar 2026 – Present

- Developed and deployed a Streamlit web application to serve real-time fragrance recommendations and dupe results
- Built a dual-model search engine combining TF-IDF and Sentence-BERT for keyword & contextual query matching
- Cleaned and preprocessed a dataset of 24,000+ fragrances using Pandas to eliminate duplicates and missing values
- Designed a 2-stage dupe detector filtering 6 budget brands from the top 50 semantic matches using cosine-similarity
- Engineered 6 note/accord columns into natural-language templates maximizing Sentence-BERT's context-awareness
- Reduced startup latency from minutes to <1s by implementing a Pickle serialization caching pipeline

NYS Food Service Compliance & Inspection Database | *SQL, RedGate Data Modeler* Mar – May 2026

- Engineered a relational database to analyze NYS food safety compliance, normalizing and ingesting 21K+ inspection records into a custom 13-table schema
- Designed a 3NF schema using RedGate Data Modeler and SQL ETL pipeline to ingest 19MB of raw data with 100% integrity, using Oracle string functions to parse complex violation strings
- Optimized SQL queries across 5+ table joins using spatial filters to extract top compliance and violation rankings

Stock Performance Dashboard | *Python, Pandas, NumPy, Matplotlib* Jun – Aug 2025

- Automated a stock performance dashboard to analyze 8 S&P 500 companies, pivoting long-format datasets to generate 27 customized Matplotlib time-series plots tracking 6 key financial metrics
- Created an automated data pipeline using the Yahoo finance API to extract and clean 1,000+ rows of historical market data over a 6-month period, eliminating manual preparation via Excel Power Query ETL

LEADERSHIP & ACTIVITIES

Leadership: Treasurer, Drexel PSA (Feb 2025 – Present) | FIFA, Host City Ambassador (Mar 2026 - Present)

Activities: Desk Assistant, Drexel University | Drexel Data Science Club | Drexel CyberDragons